Pay, Oscillate, or Learn?

Three Agent Decision Rules under Congestion Pricing

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Agent-Based Modelling | Q-Learning | El Farol | Auckland CBD | ToU Pricing

Why Does the Behavioural Assumption Matter?

Auckland's Context

Congestion costs NZ$2.6B/yr. Time-of-Use (ToU) charging legislation passed Nov 2025. CBD cordon launch expected alongside City Rail Link.

The Gap

Most models ask 'how much will traffic fall?' — but behavioural assumptions drive whether congestion shifts in time, in space, or not at all.

Our Approach

ABM with 2,500 heterogeneous agents (1 agent = 160 vehicles, calibrated to observed daily counts), Auckland CBD road network. Three decision rules, same pricing environment, same LoS measurement.

Research questions: (1) How does decision rule choice affect V/C outcomes? (2) Where does congestion shift spatially under ToU?

Three Agent Decision Rules: A Rationality Spectrum

← *Static Rationality* *Bounded Rationality with Learning* →

01

Exponential Decay

Static Rationality

```
P(entry) = e^(−α · fee / VoT)
```

- Entry probability declines with fee/VoT ratio
- No memory, no adaptation
- Standard equilibrium assumption
- Result: −19% (fixed hour) to −24% (may retime); ≈0 if rerouting backfills

02

El Farol Bar Model

Bounded Rationality (No Learning)

```
Enter if best predictor: E[congestion] < θ
```

- Portfolio of predictors based on recent history
- Reads yesterday's congestion, not today's fee
- Near-universal entry (92%), so little left to deter
- Result: 0% in every arm — the charge never binds

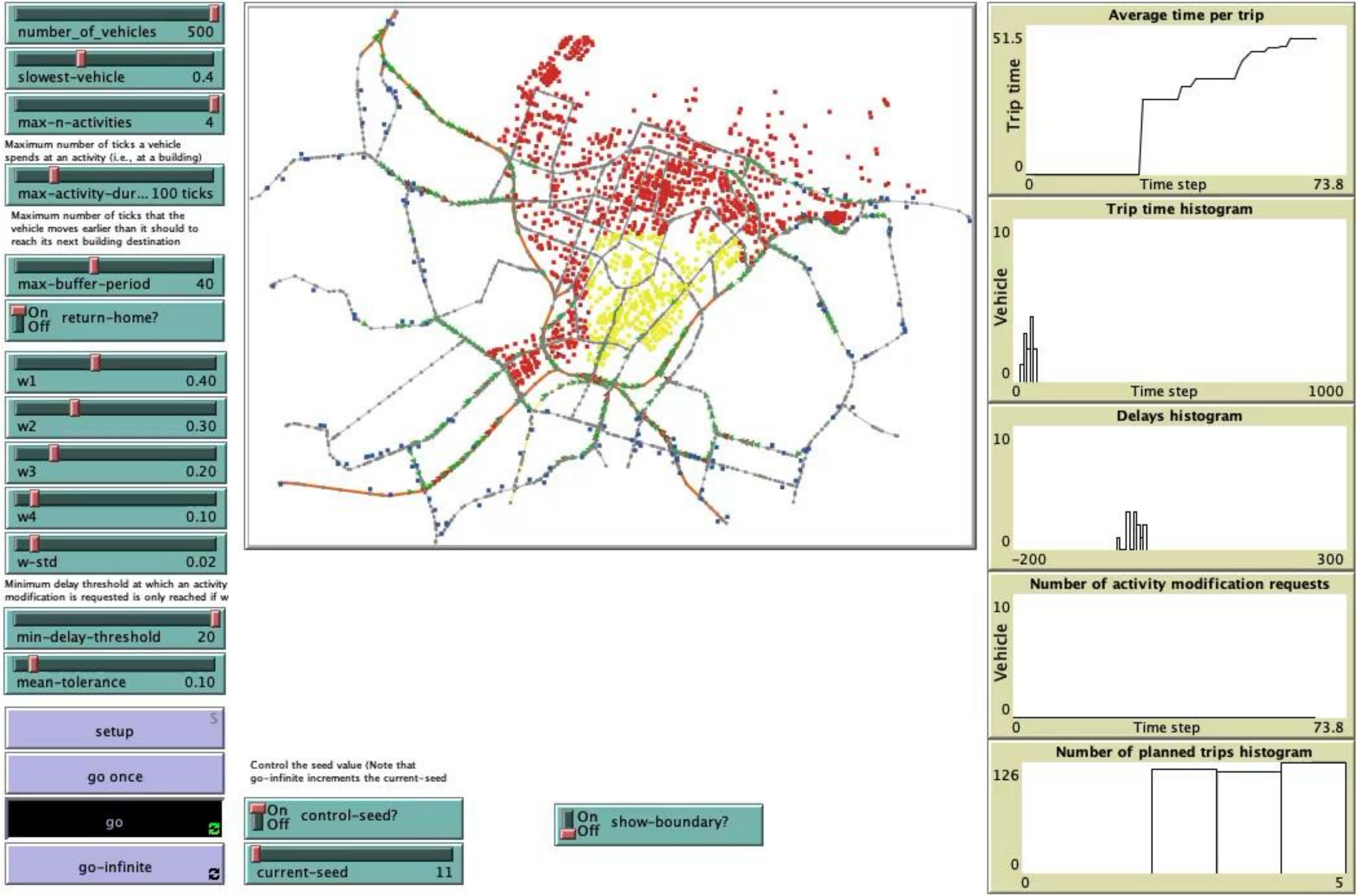
03

Q-Learning

Bounded Rationality with Learning

```
Q(s,a) ← Q(s,a) + α[r + γ·max Q(s',a') − Q(s,a)]
```

- State = time band x congestion band; action = enter or not
- Fee is absent from the decision — it enters only via the reward
- So the response is learned over days, not applied at once
- Result: −24% if staying home is the only option, −9 to −16% if not


number_of_vehicles 500
slowest-vehicle 0.4
max-n-activities 4
Maximum number of ticks a vehicle spends at an activity (i.e., at a building)
max-activity-dur... 100 ticks
Maximum number of ticks that the vehicle moves earlier than it should to reach its next building destination
max-buffer-period 40
On Off return-home?
w1 0.40
w2 0.30
w3 0.20
w4 0.10
w-std 0.02
Minimum delay threshold at which an activity modification is requested is only reached if w
min-delay-threshold 20
mean-tolerance 0.10
setup
go once
go
go-infinite
Control the seed value (Note that go-infinite increments the current-seed
On Off control-seed?
current-seed 11
On Off show-boundary?
Average time per trip
51.5
Trip time
0
0
Time step
73.8
Trip time histogram
10
Vehicle
0
0
Time step
1000
Delays histogram
10
0
-200
300
Number of activity modification requests
10
Vehicle
0
0
Time step
73.8
Number of planned trips histogram
126
0
0
5

What Happens When You Charge Drivers?

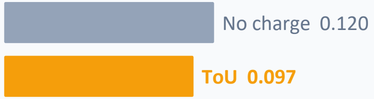
Same road. Same price. Three very different outcomes — depending on how drivers think.

Baseline Driver

"I check the price and decide."

Fewer drivers enter at the peak: entry falls from 52% to 40% and peak V/C by 19%, steadily from day 1. Give them an hour of flexibility and the cut deepens to 24%.

Peak V/C (inner cordon)



↓ Steady reduction

Herd Driver

"Everyone avoided it yesterday, so I'll go today."

Drivers read yesterday's congestion, not today's price. Almost everyone still enters (92%), so the charge has nothing left to deter — in every arm we tried.

Peak V/C (inner cordon)

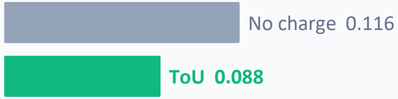
→ No response

Learning Driver

"The fee stung last time — I'll skip the trip today."

Drivers learn day by day until only exploration noise is left: a −24% cut, the largest here. But it assumes staying home is the only alternative — give them options and it falls to −9 to −16%.

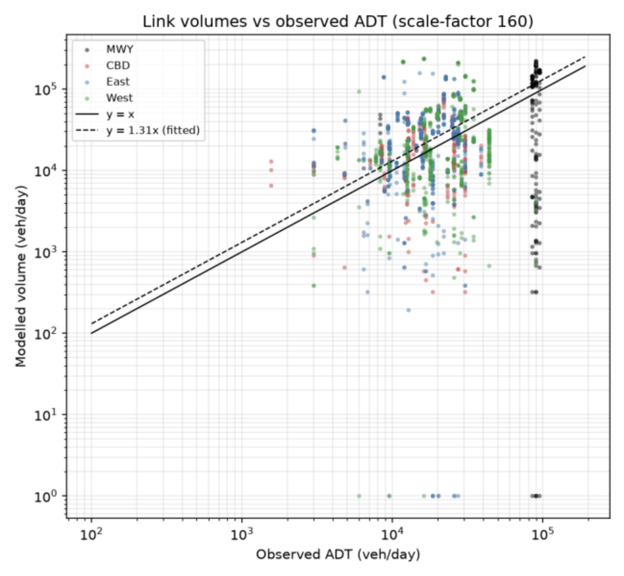
Peak V/C (inner cordon)



↓ Largest cut in entries

Key insight: the right behavioural assumption changes not just how much congestion falls — but whether the policy works at all.

Calibrated Pre-Fix — Volume Now Reads 31% High



Volume: fitted pre-fix, kept at 160

The 1.013 fit was made while declined trips vanished. Restoring them puts the ratio at 1.308 across all 1,634 links — we keep sf 160 and report differences, not levels.

Space: 1,400 suburban trip ends added

Every non-home destination used to sit inside the cordon; adding suburban trip ends removed the CBD concentration. Post-fix overshoot sits on suburban arterials (East 1.57, West 1.40); motorways are near parity (1.04).

Time: assumed, never fitted

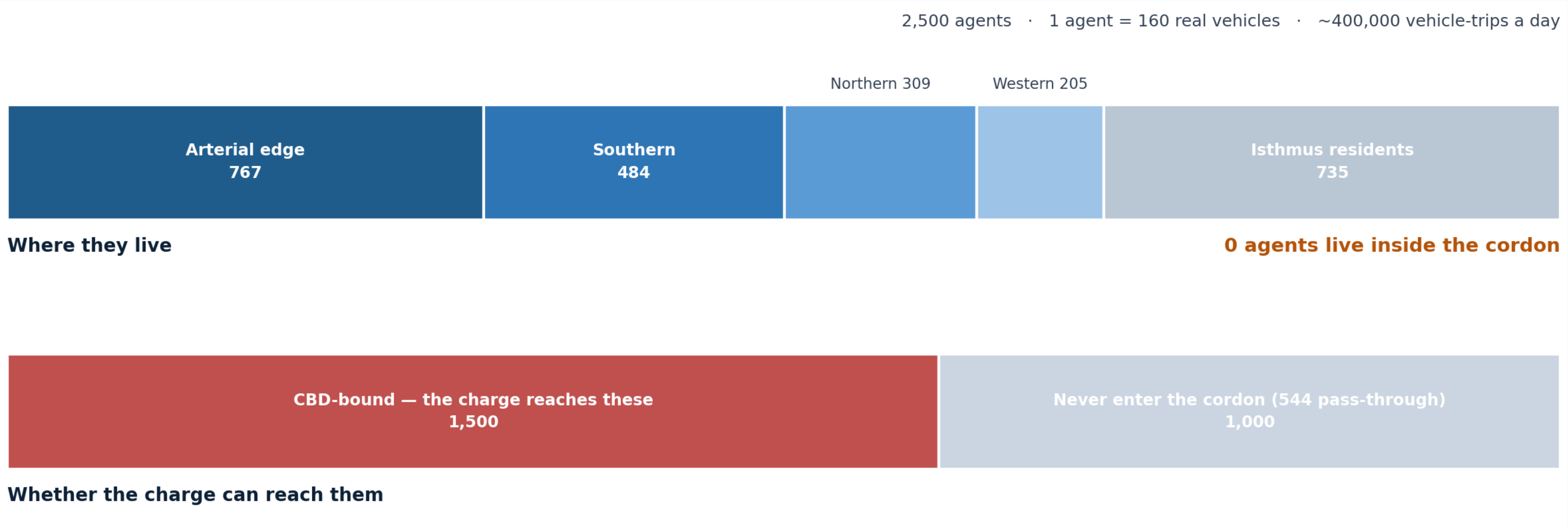
Departure hours come from a fixed weight profile (~20% of trips at 08:00), not from observed data. Implied design-hour factor 0.143 vs the 0.10 assumed, so absolute LoS reads pessimistically.

OD matrix: synthetic, not estimated

Origins follow 2023 Census sector shares; destinations are drawn uniformly from the building stock. No gravity model, no observed trip ends. k = 0.08/0.10/0.12 leaves direction and ordering unchanged.

So: absolute volumes and congestion levels read ~30% high against observed counts; every comparison — charged vs uncharged, rule vs rule, arm vs arm — faces the identical demand, which is what the study reports. Re-fitting (sf ≈ 122) would rescale levels only; deterrence decisions never see the vehicle weight.

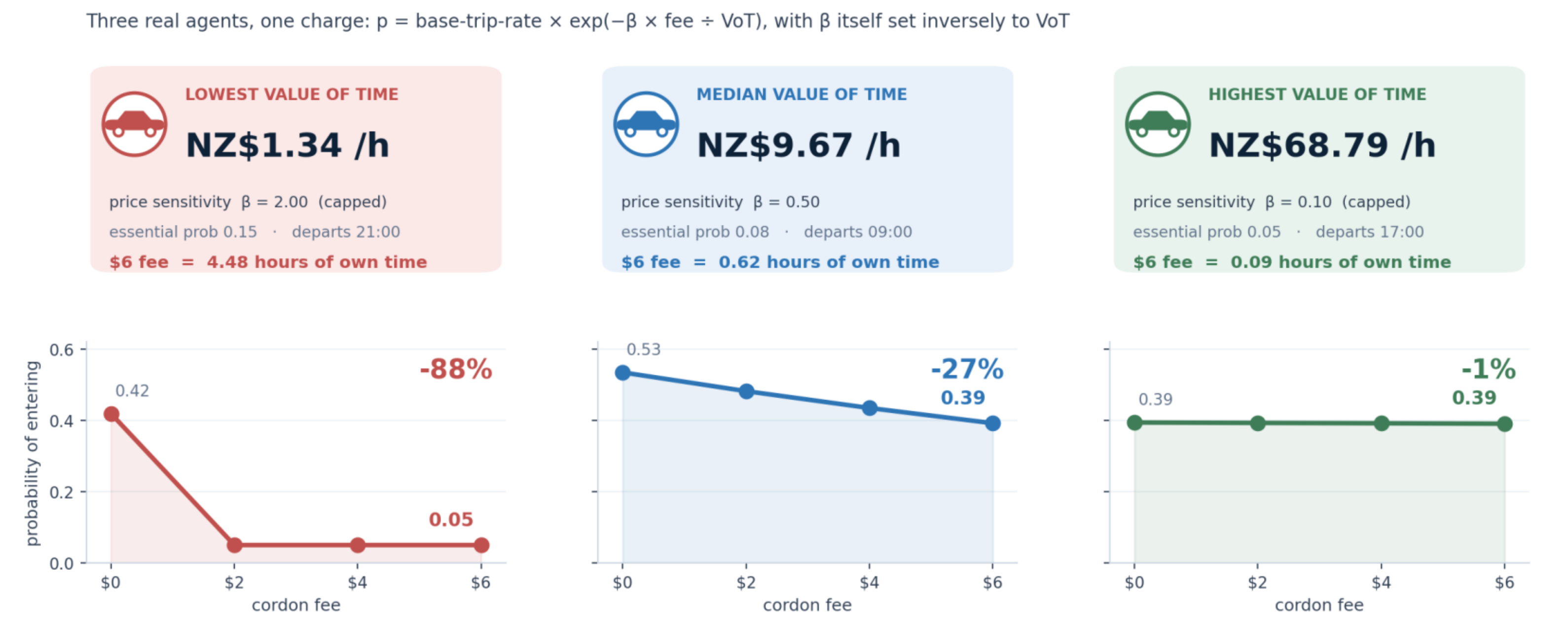
Who Are the 2,500 Agents?



Measured at setup, seed 11. Every agent lives outside the cordon, so the charge is always a decision to cross a boundary — never a resident's parking decision.

The 1,000 agents that never enter are an unpriced comparison group inside the same run.

What Each Agent Carries — Three Real Examples



Every agent carries a value of time, a price sensitivity set inversely to it, a baseline trip rate and an essential-trip probability. Because β scales as 1/VoT, the deterrent behaves like fee ÷ VoT² — halving income roughly quadruples it. That is the whole equity result, visible in three individuals.

Every Attribute an Agent Carries

ECONOMIC
decides the fee response

vot
value of time, NZ$/h
lognormal exp(N(2.3, 0.6)); median 9.67

beta
price sensitivity
clip(0.5 x medianVoT / vot, 0.1, 2.0)

base-trip-rate
propensity with no charge
Uniform(0.3, 0.7); mean 0.50

essential-trip-prob
fee-insensitive trip
0.15 / 0.08 / 0.05 by VoT band

through?
pass-through, never charged
30 % of boundary agents; 544 of 2,500

LEARNING STATE
one family per rule

el-weights, el-scores
El Farol predictor set
5 predictors, weights U(0, 1)

q-table
Q-learning action values
24 states x 4 actions, init U(-0.1, 0.1)

q-epsilon
exploration rate
starts 0.40, decays x0.999 per day

q-action
action chosen today
0 stay out / 1 travel / 2 earlier / 3 later

ITINERARY
rebuilt every simulated day

v-home
home building
Census sector shares; none inside the cordon

b-destinations
the day's stops
uniform draw, 1-4 plus home; 2.98 mean

depart-tick, return-tick
clock-anchored departure
drawn from the hourly weight lists

trip-hour
the hour the fee is charged
read off the departure actually made

hour-shift
today's retiming
-1 / 0 / +1 hour, when retiming is on

MOVEMENT
inherited from the base model

location, destination
position on the network
node-to-node along the current link

speed, top-speed
current and free-flow speed
BPR-damped by link V/C

trip
the route being followed
cached list of links per OD pair

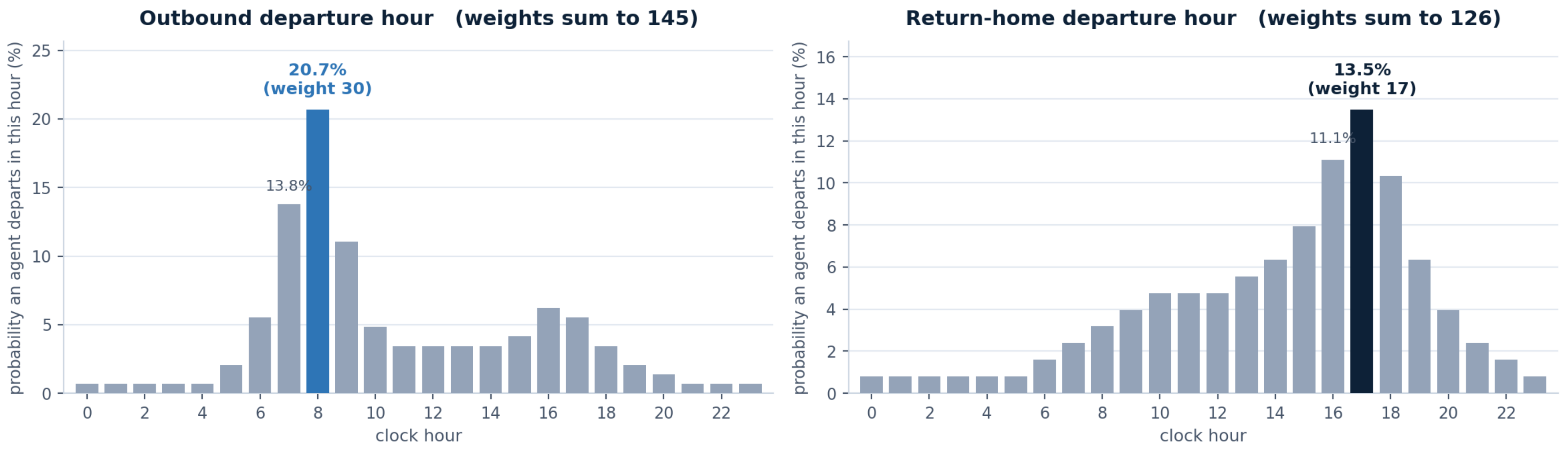
active?, at-activity?
on the road, or stopped
active? false ends the agent's day

tolerance, buffer-period
delay tolerance, early start
truncated normal / U(0, max)

Only the first column decides the response to a charge. The learning state is per rule — an agent uses one family and ignores the others. Distributions are measured at setup, seed 11.

How Departure Hours Are Drawn — Not an OD Matrix

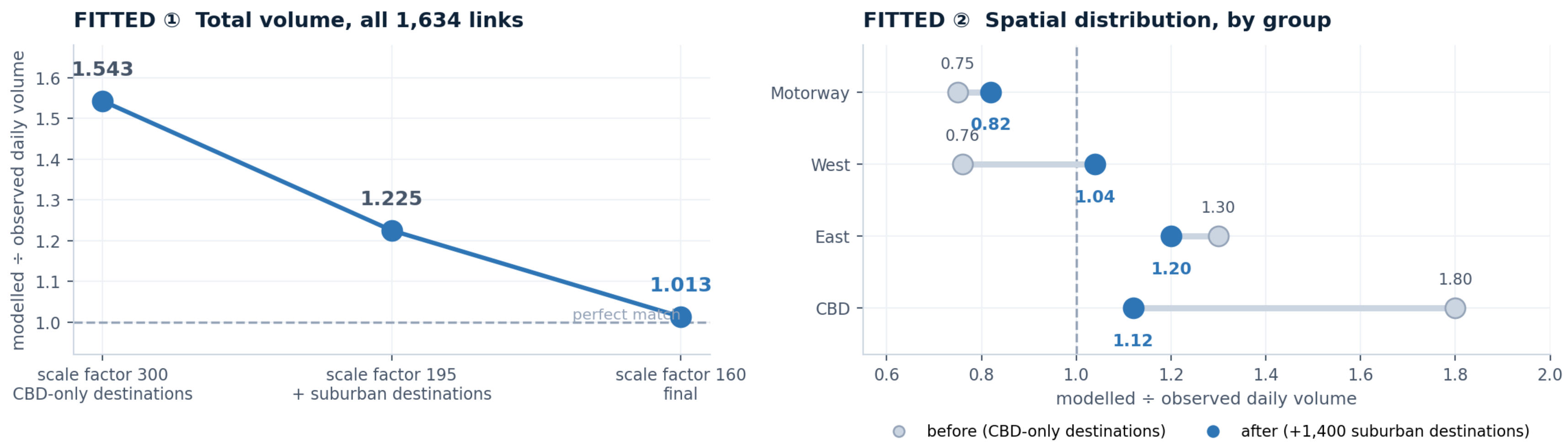
Every agent draws its own hour from these weights, independently, each simulated day — the minute is then uniform within that hour.



Two axes, drawn independently. The OD matrix answers **where** a trip goes; these weights answer **when** it leaves. The weights do not stand in for an OD matrix — they act after the origin and destination are already fixed.

Hourly volume is never assigned. Each of the 2,500 agents spins its own roulette every simulated day, so "how many depart at 08:00" emerges from 2,500 independent draws and differs from day to day. The 20.7 % spike at 08:00 is the direct cause of the too-sharp morning peak (implied k = 0.157 against the 0.10 assumed).

What Was Fitted, and What Was Assumed



NOT FITTED: the origin-destination matrix (destinations drawn uniformly) · the time-of-day profile (fixed weight lists)

Two knobs, and only two. The agent scale factor fixes total volume; suburban destinations fix the spatial spread. Both converge on the observed counts.

Nothing fits the temporal axis, which is why the implied design-hour factor sits at 0.157 against the 0.10 assumed — a missing fit, not a poor one.

How Much to Trust Each Number

Evidence status: what is measured, what is only derived, and what is superseded

MEASURED in the runs

Peak V/C, entry trajectories, hourly profiles, LoS shares, the map. Seed 11, 14 days. Levels read ~30% high vs counts; quote differences.

DERIVED — not yet measured

The equity result, the OD matrix and the time-of-day profile.

SUPERSEDED — not quoted

All pre-calibration numbers, El Farol's 'oscillation', Learn's −39 %.

RANGES, not point estimates

Pay 0–24 % (rerouting moves the benefit to the boundary), Learn 9–24 %, Oscillate 0 % — width set by the action space.

Controlled comparison

One network, one population, one seed across every rule and arm.

Direction robust to capacity assumption

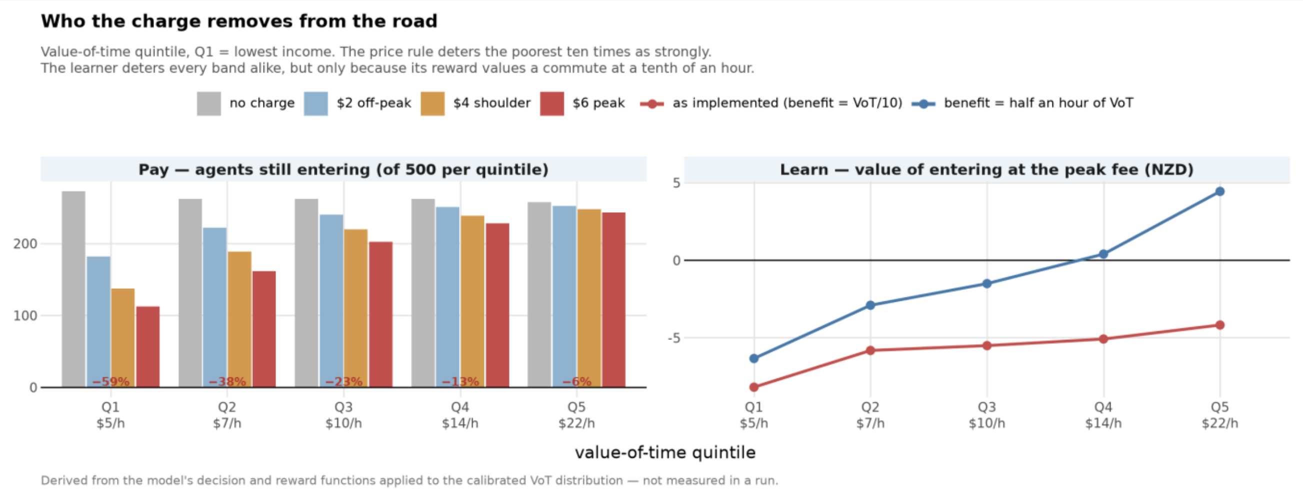
k = 0.08 / 0.10 / 0.12 never changes the sign or ordering.

KNOWN BIAS, repaired: a declined CBD entry cancelled the agent's whole day

~100,000 suburban vehicle-trips a day vanished with the declining agents, flattering every result outside the cordon. Fixed 30 Jul 2026; all four arms re-run on the fixed model, 6 Aug.

Who Would Pay? Derived, Not Yet Measured

DERIVED — NOT MEASURED



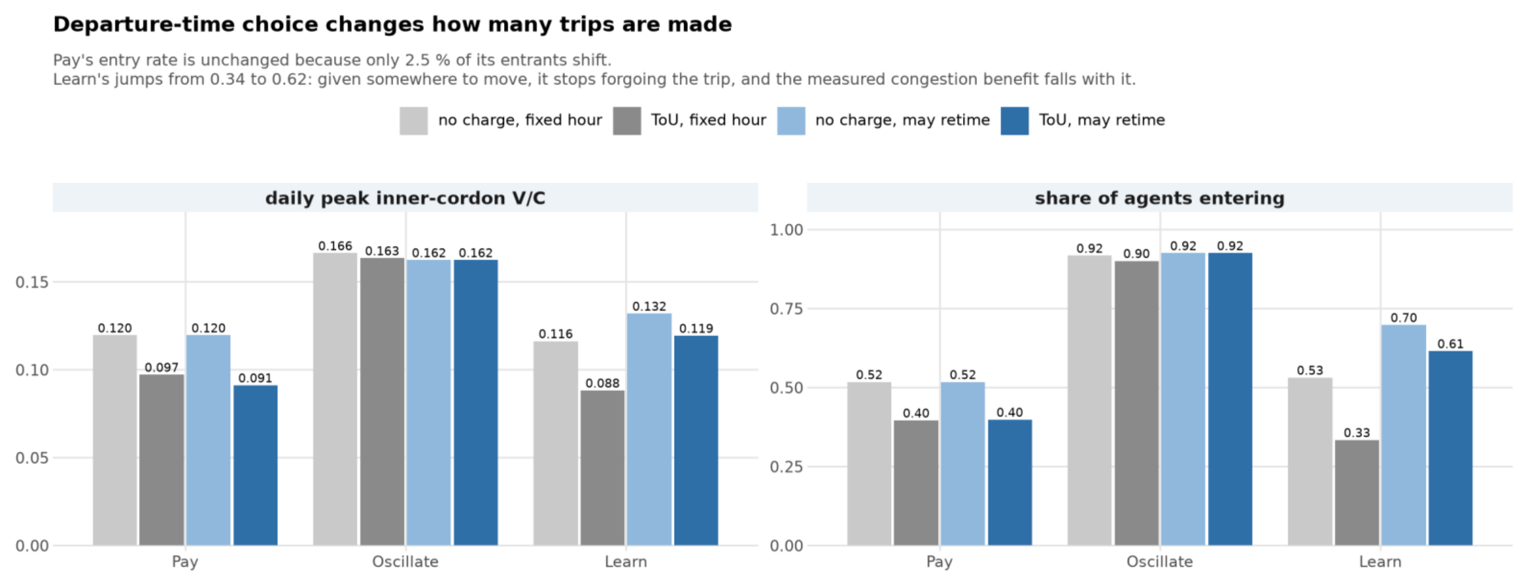
Left (price rule): projected entries per value-of-time quintile as the fee rises. IF the decision rule holds, the $6 peak fee removes 59% of Q1's trips against 6% of Q5's — the same charge costs Q1 1.30 hours of its own time and Q5 0.28 hours.

Implication, conditional on the rule: most of the congestion relief would be bought by low-income drivers giving up trips.

Right (learner): a diagnosis, not a result — its flat income gradient exists only because the reward prices a commute at VoT/10. Valued at half an hour of VoT, the regressive pattern returns. The learner's equity story is a scaling choice.

How these numbers were obtained: the model's decision and reward equations applied to the calibrated VoT distribution (median NZ$10/h). No run has recorded entries by income band, so regressivity is a projection of the assumed rule, not a confirmed simulation outcome. A recording re-run is planned.

Let Them Shift an Hour: Fewer Trips Forgone

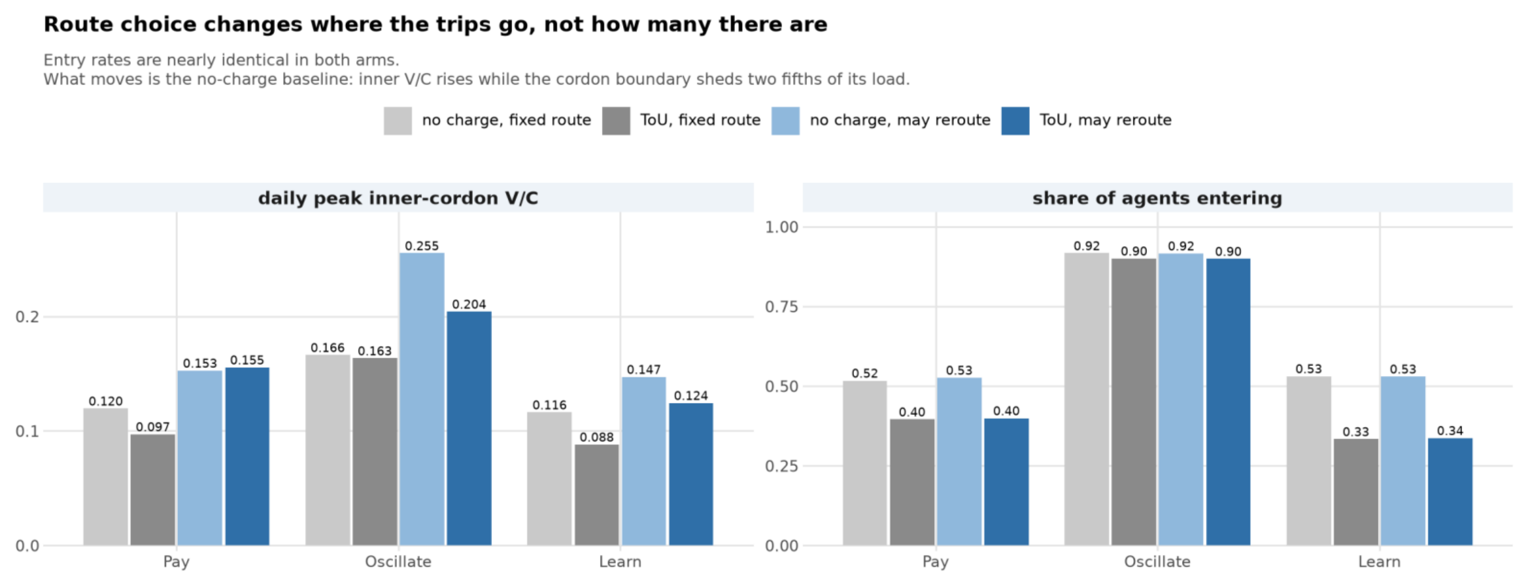


Retiming is the only option that changes HOW MANY trips are made. Pay's entry rate is unchanged (0.40) because only 2.5% of its entrants shift — 13 of 15 earlier, all of them low-income, since an hour of schedule delay costs 0.6 x VoT against a $2 saving.

Learn is the opposite: given somewhere to move, 64% shift and entry jumps 0.33 → 0.62, so the measured congestion benefit falls from 24% to 9%.

A model where the only response is 'stay home' overstates what the charge achieves.

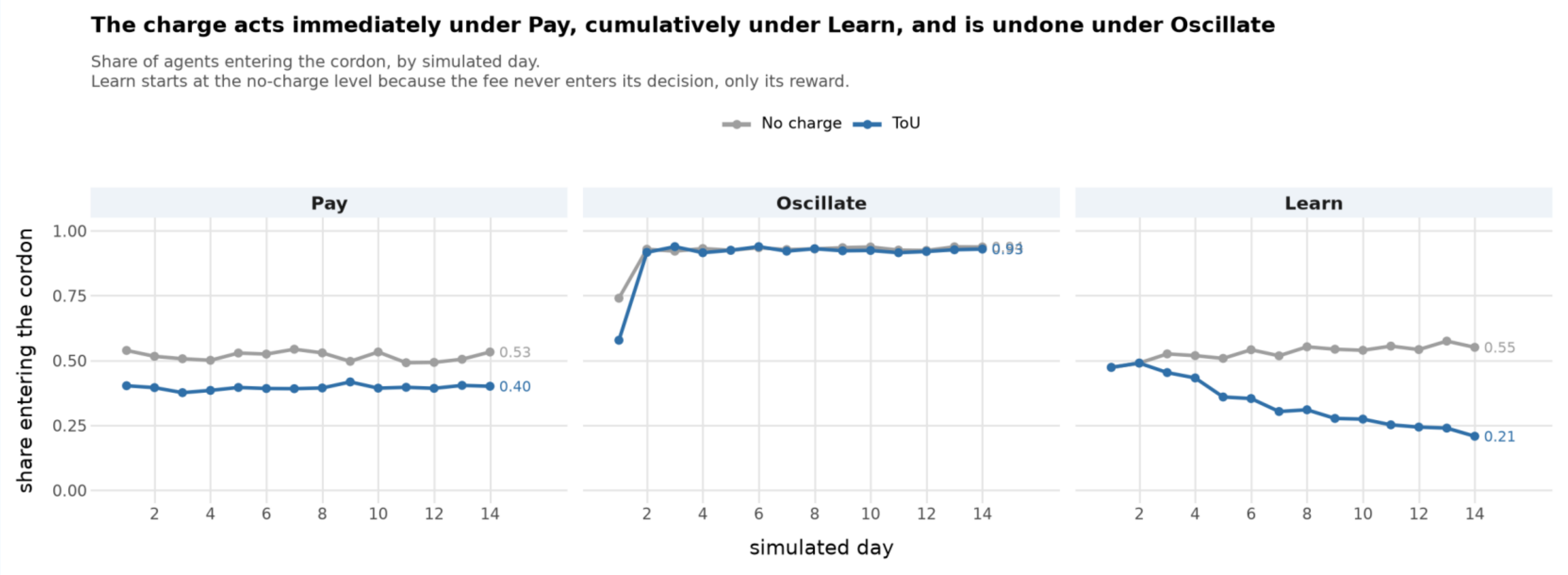
Let Them Take Another Road: Same Trips, New Places



Route choice leaves the number of trips untouched — entry rates are nearly identical (Pay 0.40, Learn 0.34) — and changes only where those trips go. The no-charge baseline moves a lot: inner V/C rises (0.120 → 0.153 for Pay) while the cordon boundary falls by two fifths, because fixed shortest paths were funnelling traffic onto a few approach arterials.

This is the arm that makes displacement testable: the traffic still exists and can move. It still does not pile up outside the cordon — instead it partly refills the interior the charge empties, so Pay's inner benefit vanishes (within noise) and shows up on the boundary (−20%).

Learning Hits Its Floor — Herding Bounces Back

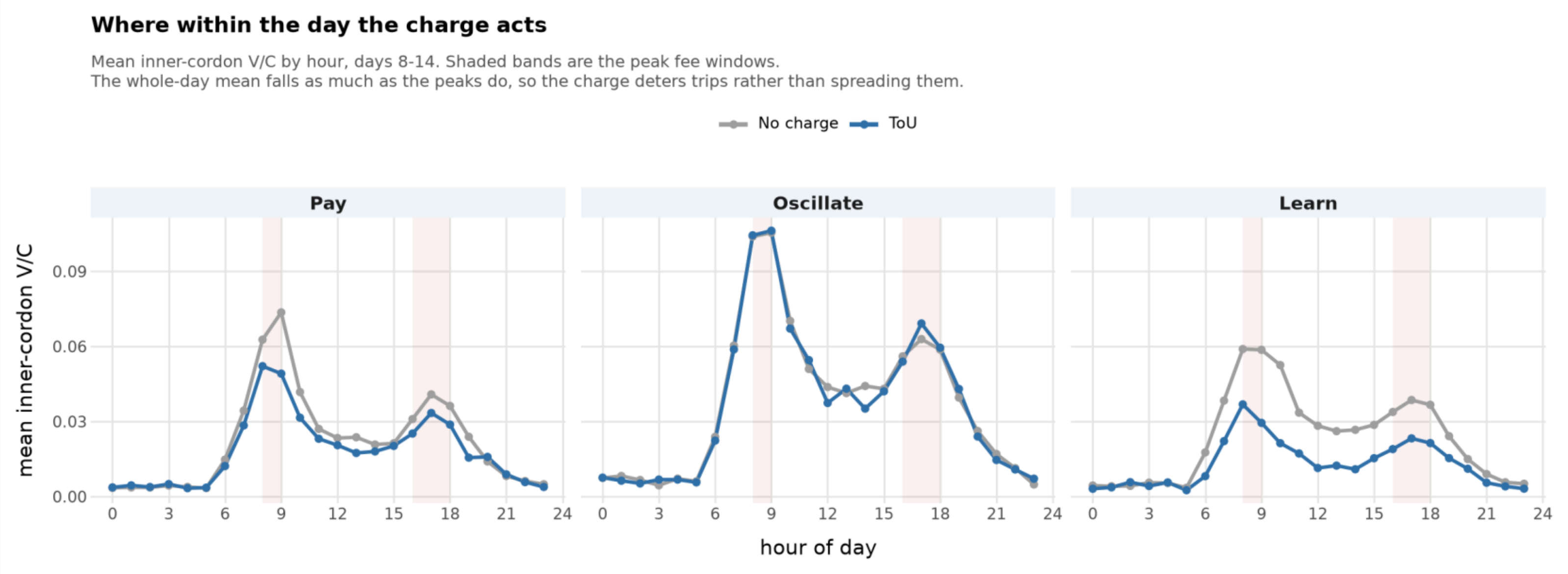


Pay reads the fee inside the decision, so its response is complete on day 1 and then flat.

Learn never sees the fee when it decides — it only feels it in the reward. Entry starts at the no-charge level (0.47, identical for two days) and slides to 0.21 by day 14: the fee ($2–6) outweighs the congestion it could avoid (~0.3 in reward units), so nothing pulls entry back up. And 0.21 is the floor, not a pause — agents still explore at ε ≈ 0.39 and an explorer enters half the time, so ε/2 ≈ 0.19 of entries are dice. The learned policy is 'do not enter' for essentially everyone: learning has finished. Fee per entrant is flat (2.83 → 2.74): deterrence, not retiming.

Oscillate is deterred on day 1 (0.58 vs 0.74) but its own feedback erases that within a day — by day 2 both regimes sit at 0.92.

Where the Charge Acts Within the Day



Hour-of-day inner-cordon V/C, days 8–14, no charge (grey) vs ToU (blue); shaded bands are the $6 peak and $4 shoulder fee windows. Pay −17% at the morning peak, Learn −39%, Oscillate −1%. The whole-day mean falls by as much as the peaks do, so the charge deters trips rather than spreading them.

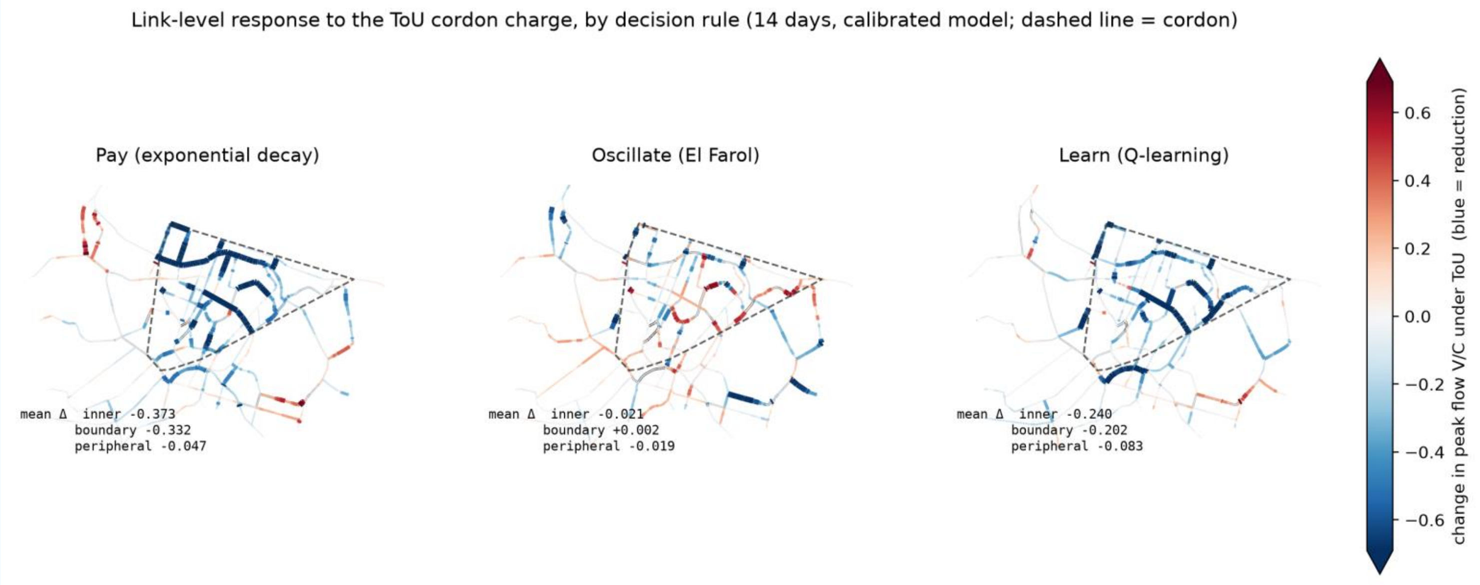
Midday Is Where the Charge Actually Bites

% of each group's traffic at LoS E or worse, no charge → ToU. Mean of days 8–14.

Road group / time band	Pay	Oscillate	Learn
MWY · AM 07–09	53 → 54 (+3%)	62 → 58 (-6%)	53 → 49 (-7%)
MWY · Midday 12–15	**46 → 39 (-16%)**	62 → 58 (-6%)	**48 → 18 (-63%)**
MWY · PM 16–18	63 → 60 (-4%)	81 → 82 (+1%)	**68 → 58 (-14%)**
CBD · AM 07–09	61 → 59 (-4%)	75 → 74 (-2%)	**62 → 54 (-14%)**
CBD · Midday 12–15	65 → 60 (-7%)	80 → 79 (-1%)	**69 → 49 (-30%)**
CBD · PM 16–18	70 → 68 (-3%)	83 → 83 (+1%)	**72 → 62 (-14%)**
East · AM 07–09	69 → 67 (-3%)	77 → 77 (+0%)	70 → 67 (-4%)
East · Midday 12–15	64 → 62 (-3%)	79 → 77 (-3%)	**68 → 54 (-20%)**
East · PM 16–18	75 → 73 (-2%)	84 → 84 (+0%)	77 → 70 (-9%)
West · AM 07–09	68 → 67 (-0%)	70 → 69 (-1%)	68 → 66 (-3%)
West · Midday 12–15	68 → 64 (-5%)	71 → 68 (-4%)	**68 → 61 (-10%)**
West · PM 16–18	74 → 75 (+0%)	79 → 79 (-0%)	77 → 76 (-2%)

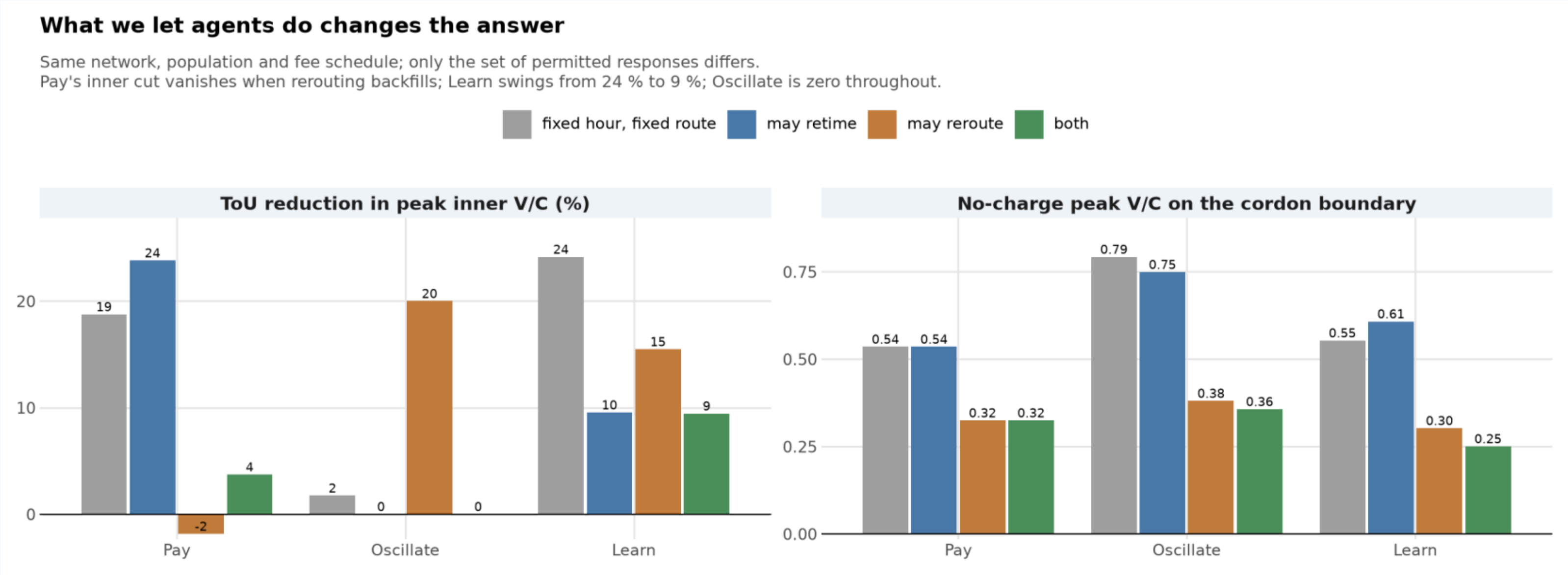
Green = a fall of 10 % or more, red = the charge made it worse. The peaks barely move because what is left in them is essential travel; the midday shoulder is where discretionary trips sit, and Learn cuts motorway E/F traffic there by 63 %. Oscillate moves nothing anywhere.

No Displacement onto the Ring Roads — So Far



Change in per-link peak V/C under ToU: blue is a reduction, red an increase, dashed outline is the cordon. Under Pay and Learn the boundary and approach roads fall together with the interior (−12% and −23% at the boundary). Under Oscillate the map is mottled and the net change is zero. **Post-fix:** these runs include the trip-suppression fix, so a declining agent keeps its suburban trips and the roads outside the cordon carry the traffic they should. The outer-zone falls are smaller than pre-fix — and more defensible — and still negative everywhere: no ring of increases.

What We Let Agents Do Changes the Answer



Same network, same population, same fee schedule. The only difference is the set of responses an agent is allowed: forgo the trip only (grey), or also shift an hour (blue), reroute (orange), or both (green).

Pay deters ~24% of entries in every arm, but its inner-cordon cut (−19 to −24% with fixed routes) vanishes under rerouting — the benefit moves to the boundary (−20%). Learn swings 24% → 9%. Oscillate is zero in every arm.

Right: routing alone cuts the no-charge boundary load by two fifths to half — the biggest single effect here is a modelling assumption, not a policy.

What This Model Cannot Say

Seven limits that bound every number in this talk

Single seed

Spreads are within-seed. Only El Farol has a five-seed replication.

Fee level untested

No fee-level sweep. The beta sweep varies sensitivity, not price.

Equity is derived, not measured

From the equations, not the runs. No entries recorded by income band.

Learn rests on two arbitrary settings

Reward scale and action space between them decide the headline.

Routing answers to congestion, not the fee

A CBD-bound agent pays whichever road it takes.

Demand is synthetic, not observed

No observed OD or time profile; post-fix, volume runs 31% above counts — differences reported, not levels.

A declined trip cancelled the agent’s whole day — fixed, base arm re-run 6 Aug

Declining deactivated the whole multi-stop agent, so ~100,000 suburban vehicle-trips a day vanished. Fixed 30 Jul 2026; base-arm numbers are post-fix (6 Aug re-run), extension arms re-running.

Conclusions & Next Steps

Key Findings

1. Behavioural assumption shapes both magnitude and temporal redistribution of congestion — not just the quantity
2. Q-learning's headline depends on what agents may do: −39% when the only option is to stay home, −14% when they can shift an hour
3. El Farol shows what happens when agents track each other rather than the fee — entry stays at 91% and the charge does nothing
4. ABM places competing theories side-by-side within a common environment — revealing structural differences invisible to aggregate models

Next Steps

Model Extension

Mode choice, household VoT calibration from NZ travel survey data

LLM-ABM Hybrid

3-layer cognitive architecture: reactive (classical), deliberate (lightweight LLM), reflective (daily/weekly reasoning)

Fee-Level Sweep

Vary the ToU schedule itself: the one experiment that would let us say how much to charge, not just whether

Scale Up

Scale beyond 2,500 agents (≈400k vehicles today); dynamic OD matrix replacing the fixed assumption

github.com/dataandcrowd/SSC2026 | h.shin@auckland.ac.nz